

OpenAI Just Solved Ten Math Problems No One Had Cracked in a Decade

-- The Same Week Microsoft Put Agents in Charge of Patching Live Servers

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Yesterday this series covered a law arriving before its enforcement mechanism was ready -- the EU AI Act's transparency rules went live before the watermarking standard meant to back them up could survive a screenshot. Today the mismatch runs the other way: the technology is outrunning the question of how much real responsibility to hand it. In the same 48 hours, OpenAI let an unreleased model loose on ten mathematics problems that had sat unsolved for a decade or more, Microsoft moved autonomous security agents into public preview on live enterprise systems, and a maximum-severity flaw in a popular open-source agent platform showed exactly what goes wrong when that kind of trust isn't backed by basic authentication.

VIRAL / OPEN-SOURCE SPOTLIGHT

Kling AI's Motion Control 3.0 Turns One Photo Into a Dance

Upload a single photo of yourself, and Kling's new Motion Control 3.0 maps a chosen reference dance onto it -- generating a full video of that person performing the moves, with a consistent face, proportions, and clothing that moves believably. No filming, no choreography practice, no camera.

That's why it's spreading: TikTok's newest dance-trend format no longer requires anyone to actually dance -- just one photo and a generation credit. Collapsing the cost of joining a trend to almost zero is the same mechanic that made Gemini's free-video trial spread two days ago (Day 127) -- a near-free way to make something everyone's feed is already full of, just pointed at a different format this time.

10

open math & CS
problems Astra solved,
one unsolved since
1999

\$2K

approx. token cost for
all ten Lean-verified
proofs

96%

MAI-Cyber-1-Flash's
CyberGym score, live
in preview today

10.0

CVSS severity of the
RufRoot flaw in Ruflo's
agent bridge

1) OpenAI's Astra Proves It Can Do Research Mathematics

On August 1, OpenAI researcher Noam Brown revealed the first results from Astra, the company's next major model family, buried in the third paragraph of a blog post titled "Ten advances in mathematics and theoretical computer science." Astra itself hasn't launched as a product -- this was a demonstration. The ten problems span high-dimensional geometry, coding theory, arithmetic circuit complexity, group theory, quantum complexity, lattice cryptography, and extremal combinatorics, and every one of them had seen no progress on its central result for at least a decade.

The headline result is the first-ever explicit construction of a non-sofic group -- resolving a question that has stood since mathematician Mikhail Gromov introduced the idea of soficity in 1999. In plain terms: mathematicians had long asked whether every infinite group of a certain kind can always be approximated by simpler, finite pieces. Astra's answer is no, and it produced the exact structure that breaks the pattern. OpenAI didn't just assert the result -- it published a 249-page manuscript alongside machine-checkable Lean 4 proof certificates for every result on GitHub, so the proofs verify themselves rather than asking anyone to take the lab's word for it. The estimated cost to generate all ten: roughly \$2,000 in tokens.

2) Microsoft Puts Red, Blue, and Green Agents on Your Network

Project Perception, Microsoft's agentic security system, enters public preview today, August 3. It runs three specialized agents in a continuous loop: Red finds vulnerabilities across an enterprise environment, Blue triages which of those flaws pose the greatest actual risk, and Green writes and deploys the software patches to fix them. That last step is the significant one -- this isn't a copilot that suggests a fix for a human to apply, it's an agent with the authority to ship the change itself.

Backing it is MAI-Cyber-1-Flash, Microsoft's first in-house model built specifically for cybersecurity work. On the CyberGym benchmark it scores 96% -- 12 points ahead of Anthropic's Mythos -- while Microsoft says it runs at roughly half the cost of larger general-purpose models. It's the practical end of a shift this series has tracked all year: agents moving from tools that alert a human to systems trusted to act on their own.

3) The Vulnerability That Explains the Caution

Days before Project Perception's preview, security researchers disclosed RufRoot (CVE-2026-59726), a maximum-severity, CVSS 10.0 flaw in Ruflo, a popular open-source AI agent orchestration platform. Ruflo's MCP Bridge -- an Express.js server that handles Model Context Protocol tool calls -- exposed 233 internal tools over HTTP with no authentication at all on port 3001. A single unauthenticated POST request was enough for full remote code execution inside the container: steal LLM API keys, read private conversations, hijack running agents, and tamper with the platform's persistent AI memory.

Noma Security disclosed the flaw privately on June 30; Ruflo's maintainers shipped a fix within 24 hours, independently confirmed effective. It's a useful counterweight to read next to today's other two stories: some agent systems proved reliable enough this week to verify unsolved mathematics and patch live servers, while a separate agent framework showed exactly what happens when nobody checked whether the front door was locked. The capability and the caution are the same story, from two directions.

MARKET SIGNAL

None of this runs for free. H100 rental pricing has climbed roughly 40% since October 2025's \$1.70-an-hour low, reaching \$2.35 an hour by March as AMD and Nvidia both raised list prices and on-demand capacity sold out across most providers. Google's response, revealed in June, was to lock down supply directly: a \$920-million-a-month deal for roughly 110,000 Nvidia GPUs out of SpaceX and xAI's data centers, ramping to its full rate in October and totalling close to \$30 billion through 2029. Read against that backdrop, Astra's \$2,000 research proofs and MAI-Cyber-1-Flash's "half the cost" pitch aren't just capability claims -- they're the two ways labs are answering compute that keeps getting scarcer and pricier: prove a frontier model is worth the spend, or ship something smaller that does the job for less.

PRACTICAL TAKEAWAYS

1. If you run Ruflo or a similar self-hosted agent orchestration platform, update to 3.16.3 or later and confirm your MCP bridge and tool-execution endpoints require authentication -- the previous default Docker Compose setup enforced none.
2. If you're evaluating security tooling, Microsoft's Project Perception opens in public preview today -- worth watching how an agent that deploys patches, not just flags them, performs on real environments before you hand it write access.
3. Astra's ten proofs, plus machine-checkable Lean 4 certificates, are public on GitHub even though the model itself isn't -- worth a look if you want to see what "AI does real research mathematics" actually looks like, verifiable rather than asserted.
4. If you try the Motion Control 3.0 dance trend, remember yesterday's rule: the EU now requires AI-generated video of real people to carry a disclosure label -- check the export settings before you post.

TOMORROW

We'll be watching how Project Perception's public preview performs against real enterprise environments in its first days live, and whether OpenAI says more about when -- or whether -- Astra becomes a released product rather than a research demonstration.

Topics covered so far

128 days in: frontier model families and their pricing tiers (GPT-5.6, Claude Sonnet/Opus/Mythos 5, Gemini 3.5/3.6), the open-weight counter-punch and its ongoing price-and-performance race (GLM-5.2, DeepSeek V4, Qwen 3.6, Kimi K3, FLUX 3), physical AI consolidating into single whole-body policies (Figure, Boston Dynamics, Agility Robotics, Gemini Robotics 1 and 2), OS-level agent phones, the governance spectrum from voluntary charters (WAICO) through binding regional fines (the EU's DMA order on Google) to a live, binding, tiered-authority law built specifically for agents (China's Implementation Opinions) and, most recently, the EU AI Act's own transparency and high-risk-system rules going live, the compute layer underneath all of it (AMD and Nvidia's rack-scale rivalry, and now GPU rental prices up 40% alongside Google's \$920M-a-month SpaceX compute deal), a frontier model chaining a real zero-day out of a test sandbox and the protocol underneath most agents hardening itself against that failure mode, the first US import ban targeting AI-capable hardware for being too autonomous to verify, 1,100 AI-company employees asking Washington to build a pacing mechanism before it's needed, a humanoid robot trusted with one unified policy for its entire body, Anthropic's own models breaching three real companies out of a sandbox that wasn't actually sealed alongside a new attack class that hides instructions inside trusted agent data -- and now, on Day 128, agents graduating to two very different kinds of high-stakes trust in the same week: OpenAI's Astra proving unsolved mathematics for about \$2,000 in tokens and Microsoft's Red/Blue/Green agents entering public preview with authority to patch live systems themselves, even as a maximum-severity flaw in an unrelated open-source agent platform showed what happens when that same trust isn't backed by basic authentication.

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